

Risk and Uncertainty in Economic Evaluation of Medical Care Technology

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Introduction

Resources—people, facilities, equipment, time, and knowledge—are limited but the demand for health care is unlimited. Therefore, decisions have to be made in allocating limited resources across competing health care interventions. “Economic evaluation is the comparative analysis of alternative courses of actions in terms of both their costs and consequences”.¹ Therefore, the basic steps of any economic evaluation are to identify, measure, value, and compare costs and benefits of the alternatives being considered. Economic evaluation of health care technology is crucial to promote economic efficiency in the allocation of scarce resources of health sector.²

The logic of economic efficiency was discussed in the writings of Vilfredo Pareto, Nicholas Kaldor and Sir John Hicks. A separate branch of economics called welfare economics was developed by the end of World War II to define the optimal allocation of a society's scarce resources.

Many economic decisions are made under the conditions of certainty where consumers know fairly well the present and future prices, incomes, and other economic parameters. However, there is a substantial amount of risk or uncertainty regarding future outcomes for many economic decisions. The difference between risk and uncertainty is that in the former we can list all outcomes and assign probabilities to them but in the later situations we may neither be able to list the outcomes nor assign the probabilities.³ Theory of individual choice under uncertainty had been developed on the basis of expected utility theory to tackle uncertain economic situations.

In his classic article “Uncertainty and the welfare economics of medical care”, Arrow (1963) argues that... “the special economic problems of medical care can be explained as adaptations to the existence of uncertainty in the incidence of disease and the efficacy of treatment”.⁴

What are the special economic problems of medical care?

- (1) The nature of demand in the medical care market is irregular and unpredictable.
- (2) The incidence of disease, that is, diagnosis and prognosis are not cent percent perfect.
- (3) Many illnesses occur at random and when they occur, they entail substantial medical costs. Everybody have some probability of becoming ill or sick.
- (4) In extreme, many illnesses associate with discomfort (death) and loss of earnings.
- (5) There is also risk regarding recovery or effectiveness of care.
- (6) In addition, information asymmetry exists between doctor and patient in the medical care market. Here, doctor is informed and patient is uninformed player. Information asymmetry exists between these two players on the information of effectiveness of medical care.

Every economic evaluation study of medical care technology or program involves a lot of uncertainty. To conduct a complete economic evaluation study, the analyst combines the information on the course of disease and treatment, the clinical effectiveness of regimens, preferences regarding health outcomes, the costs of the intervention and other aspects of clinical problem.⁵ Results of randomised clinical trials and observational studies provide many of these pieces of information. However, each piece of information is subject to a wide range of uncertainty. The aim of this paper is to review the current methods of dealing with uncertainty in economic evaluation of medical care technology. A brief discussion on the sources of risk and uncertainty is presented in section two. Section three reviews the techniques those are recently used to estimate and handle the risk and uncertainty. Finally, a conclusion is provided in section four.

Sources of Uncertainty

There are a number of sources of uncertainty in economic evaluation.

First, in some cases, data for economic evaluation may not be available. Therefore, it is necessary to make informed guesses. For example, effectiveness data of new medical technology are not available.

Secondly, data may be available but they may be known to be imprecise. For example, in case of the estimates of hospital costs only the average cost per day, or per admission, is known.

Thirdly, some key parameter can not be known because we have not observed it or could not observe it. For example, the future rate of medical inflation relative to other goods and services can not be known before it occurs.

Fourthly, value judgement for the choice of appropriate rate of some key parameters may create uncertainty. For example, the appropriate rate of discount for social decisions is a problem of this type.

Fifthly, methodological controversy is an important source of uncertainty of economic evaluation study. For example, the source of values for health state preferences or inclusion/exclusion productivity changes in the analysis create uncertainty.

Finally, the epidemiology of the disease, patterns of physician behaviour and patient compliance may be uncertain.

In fact, we want to know how uncertain our analysis or the result is. The less uncertain the result is the more confidence one can place on it. Therefore, it is essential to provide some assessment of uncertainty in the analysis.

Methods of Dealing with Risk and Uncertainty

Until recently, sensitivity analysis has been the main method of dealing with uncertainty in economic evaluations of medical care technology. Moreover, classical statistical measures of uncertainty have been used because of the increasing practice of collecting economic data alongside clinical trials. In addition, the Bayesian inference approach is becoming popular in economic evaluation study.

Sensitivity Analysis

Sensitivity analysis is very crucial for a sound cost-effectiveness analysis (CEA). This method is widely used to assess the problems of data uncertainty in economic evaluations of medical care programs. Program like Lotus 123 or Excel can be used to carry out sensitivity analysis in CEA.

Sensitivity analysis is a way of acknowledging uncertainty about the values of important parameters in our predictions. The estimate we obtained based on the assumptions of key parameters is called base case estimate. The aim of sensitivity analysis is to examine the robustness of base case estimate over a range of alternative values for uncertain parameters.⁶ If the major results do not change when we consider the range of reasonable assumptions, then our analysis is robust and we can have greater confidence in its results. If the results are sensitive to which in a range of plausible values of a parameter is used, then the conclusions are not robust. Policy makers need to be very careful over the range at which the results are sensitive. In such cases, more precise data are required to draw a definitive conclusion. The beauty of a sensitivity analysis is that it focuses our attention to critical values of the uncertain parameters. It also creates a foundation of analysis of whether or not it is

worthwhile to generate additional information on uncertain parameters for policy conclusions.

Univariate Sensitivity Analysis

The traditional approach to sensitivity analysis is to examine one variable at a time. However, the point is the selection of the range of uncertain parameter over which sensitivity analysis will be carried out. In some cases a high and low value as well as a "best" estimate are selected. In other studies, 95% confidence intervals for key parameters are used to determine a plausible range for variation. Another practice is to vary the uncertain parameters by plus or minus one standard deviation. Finally, some have advocated the use of a clinically meaning range.

Multivariate Sensitivity Analysis

The CEA is based on many parameters. For example, costs depend on the discounted sum of physician fees, hospitalisation costs, drug costs, and others. Thus, examining one source of uncertainty at a time grossly under estimate or over estimate the overall uncertainty in the estimate of the cost-effectiveness ratio.

There are several ways to deal with multiple sources of uncertainty.

Firstly, we can vary multiple parameters at a time.

Secondly, we can combine uncertain parameter values to obtain the worst and the best cost-effectiveness ratios.

Finally, Monte Carlo Sensitivity Analysis is a very sophisticated way of dealing with uncertainty that overcomes the problems associated with univariate and extreme case sensitivity analyses.⁷ For example, values near the base case assumptions are more likely to occur than values near the extremes of the range. To perform Monte Carlo Analysis, we need to specify probability distributions for all the important uncertain quantitative assumptions. We then execute a trial by taking a random draw from the distribution for each parameter to arrive at a set of specific values for computing realised cost-effectiveness ratios. We repeat the trial many times to produce a large number of realisations of cost-effectiveness ratios. The average of the trials provides an estimate of the expected value of cost-effectiveness ratios.

Potential Parameters/Assumptions for Sensitivity Analysis

Sensitivity analysis can be used to address the valuation of nonmonetized outcomes. For example, some measures of health related quality of life are still to be tested for validity, consistency and reliability and the health outcomes obtained from these measures should be included into sensitivity analysis. Alternative modelling assumptions have potentials for sensitivity analysis.

The important parameter that requires sensitivity analysis is the discount rate because there is no consensus among the economists on a suitable discount rate to carry out CEA.

Limitations of Sensitivity Analysis

The weaknesses of sensitivity analysis are as follows:

Firstly, the analyst has the full discretion to choose which variables to vary and which to treat as known or fixed. Therefore, it creates selection bias.

Secondly, the analyst determines the amount of variation around the base value of the parameter. In most cases, variables are varied by randomly selected values. Moreover, there are no guidelines about the acceptable range over which the conclusion of the study is robust.

Finally, one way sensitivity analysis fails to incorporate the interactions among the key parameters. Since net benefit or cost-effectiveness ratio is a random number, one must consider multiple sources of uncertainty once at a time. Failing to do so, of course, will create bias in the estimation of overall uncertainty.

Expected Value Analysis

Expected utility theory developed by Von Neumann and Morgenstern provides us a strong theoretical foundation to make decisions under risky situation. A simple example is presented to explain the theory. A single disease is considered, and a patient either has the disease or does not. An effective treatment is available. The outcome of the treatment can be measured in terms of either additional life years gained or utilities or monetary units. For easy of understanding, let's assume net benefits are expressed in money terms. The payoffs with and without treatment are shown in the following table:

Table 1: Payoffs with and without treatment

Events	Acts	
	Treatment	No Treatment
Disease	Tk.80	Tk.40
Healthy	Tk.90	Tk.100

For example, if the patient does not have the disease and does not treated, a net benefit of Tk.100 will be achieved, but if the patient has the disease and is treated, he will have a net benefit of Tk.80.

Suppose, we believe that the prior probability of disease is 0.5 so the prior probability of no disease is 0.5. Now, we can calculate expected value of each action.

$$EV (\text{Treatment}) = (80)(0.5) + (90)(0.5) = 85$$

$$EV (\text{No Treatment}) = (40)(0.5) + (100)(0.5) = 70$$

If our aim is to maximise patient's expected net benefit, it is recommended to choose the treatment action, as this has the higher expected net value.

However, the distribution of outcomes i.e., risk is also important in making decisions. Expected value criterion fails to incorporate this into analysis. For example, let's assume the outcomes of no treatment option are same i.e., Tk.85 whatever the event is. The expected net value of no treatment will be Tk.85. In this case, patient would be indifferent between treatment and no treatment according to expected value criterion but he would not. He would rather prefer no treatment because a certain outcome of Tk.85 would occur in this action. A risk averse individual would prefer an outcome with certainty to a gamble that has the same expected value.

If the patient were indifferent between having Tk.83 with certainty, and having a fifty-fifty probability of either 80 or 90, then the certainty equivalent of treatment action would be 83. The patient is willing to give up Tk.2 in the expected value to have the certainty equivalent. Therefore, the cost of risk (K) would be 2.

Suppose, the costs of a project incurred in the initial year and benefits would occur only in the following year. Therefore, the net present value (NPV) could be expressed in the following way:

$$NPV = -C_0 + B_1/(1+i)$$

Where,

C_0 = Costs incurred in the initial year

B_1 = Benefits would occur in the first year

i = Discount rate

Since future benefits are uncertain, they need to be discounted at the appropriate riskless discount rate. We get the following formula by incorporating risk premium.

$$NPV = -C_0 + B_1/(1+i+\rho)$$

Where,

C_0 = Costs incurred in the initial year

B_1 = Benefits in the first year

ρ = Risk premium

As we know how to estimate the cost of risk, we can adjust the criterion. The correct method is called present certainty equivalent value (PCEV) and can be calculated by the following formula:

$$PCEV = -C_0 + (B - K) / (1 + i)$$

Where,

B = Expected Benefits

As a result, it is not required to adjust risk in the discount rate. However, one needs to be very careful to determine risk because subjective valuation will be biased. Appropriate empirical data should be employed to estimate risk. Another limitation of this method is that it implicitly assumes greater risk for long-gestating projects.

Statistical Methods

Economic evaluation alongside clinical trials has created another possible source of uncertainty that may arise due to sampling variation. Several statistical methods have been developed to handle this sort of uncertainty. Any textbook of Statistics provides a detailed analysis of these methods. The comprehensive discussion of each method is beyond the scope of this paper. Here, we just give a brief discussion of these approaches. The delta method is frequently used to estimate the uncertainty in the cost-effectiveness ratio. This is just an application of a Second-order Taylor series expansion to the estimation of the variance of some function. The method could be used to calculate the variability of cost-effectiveness ratio or any component of costs or effectiveness. However, the delta method may provide biased estimate of uncertainty. Another alternative way is to estimate the parameters to simulate a distribution of cost-effectiveness ratio by repeatedly taking draws. The resulting distribution provides a consistent estimate of cost-effectiveness ratio. This approach is similar to Monte Carlo Sensitivity Analysis. Finally, a bootstrap method can be applied to estimate the distribution of cost-effectiveness ratio. Under this method, one draws a sample from all observations at random and estimates a C/E ratio. This process is repeated many times to get a distribution of C/E ratios. The method can only be applied if the data on all the variables are available for the analysis. The

beauty of this method is that it can capture multiple sources of uncertainty once at a time and is suggested if sufficient data is available.

The Bayesian Approach

The Bayesian approach of estimating uncertainty is the counterpart of the "Classical" Statistical approach. According to the classical approach, for example, a parameter is only a function of observed data while according to the Bayesian approach, it is not only a function of observed data but also a function of the probability distribution of the possible values of the parameter. The Bayesian approach strongly believes that prior probability distribution has an effect and that should be incorporated along with observed data to estimate a parameter.

Conclusion

Uncertainty is a fact of life. Medical care market is not excluded from it. Economic methods have been successfully applied to deal with risk and uncertainty in economic evaluation of medical care technology. If the findings of the study are found to be very sensitive to an uncertain parameter then one needs to collect more information about that uncertain parameter before taking any decision. If the information is available to derive an unbiased estimate of an uncertain parameter then no problem exists to take a reasonable decision. However, if information regarding uncertain parameter is not available, then there exists an additional option- wait and look for better information. This option of deferring decision is particularly important when investment in new health care technology, at first stage, is huge. Irreversible nature of investment in many health care technologies makes this option very crucial to incorporate in economic evaluation of such technology.⁸ Using decision analytic approach, it has been shown that eliminating any feasible alternatives by intuitive judgements may be misleading and a potentially cost-effective alternative may be ruled out prior to an economic evaluation study.⁹ It has also been shown that current practice may not be a relevant alternative to evaluate a new health care technology because for mutually exclusive programmes decisions have to be made on the basis of mean net benefits whether it is measured on effectiveness or monetary scale and classical or Bayesian inference is irrelevant to make clinical decision because these inferences can seriously damage our health.¹⁰ Both type of inferences or perhaps Bayesian inference are very relevant to further evaluate whether the current practice is cost effective or not by estimating the value of information. However, for new health care technology which involves big investment and subjects to irreversibility and indivisibility, deferring option is essentially an important relevant alternative and the importance of this deferring option is further stimulated by the rapid innovation of new technologies in this sector. Traditional decision rules of economic evaluation may not be consistent for such type of technologies even if we could explicitly establish a critical cost-effectiveness ratio or a monetary value of health outcome.¹¹ Existing methods of dealing with uncertainty may not also be applicable in this

circumstance. An alternative approach developed by Palmer and Smith may be relevant to quantify this sort of uncertainty. This alternative approach is derived with some modification from 'Option Pricing' technique that is frequently used in commercial and business sector. They have shown that the presence of even modest degrees of uncertainty can give rise to substantial changes in the investment criterion for economic evaluation. Of course, the relevance of option pricing technique in economic evaluation depends on the particular characteristics of health care technologies under consideration.

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